20250303EB_MN(GC-MS)

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2025年3月3日 星期一

Use Neurobasal-A without glucose and sodium pyruvate for experiment. I didn't add extra Na Pyruvate this time.

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年3月5日 星期三

T2 -> T3

2025年3月7日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年3月10日 星期一

T4 -> T5

2025年3月12日 星期三

T5 -> T6

2025年3月14日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年3月17日 星期一

T6 -> T7
Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

Table3

	A	B	C	D
1	conc. (M/mL)	survival	total (5mL)	M/well (into 9*6-well)
2	2.07	100	10.35	1.15

20250310 coat

<table>
<tr><th></th><th>1</th><th>2</th><th>3</th></tr>
<tr><td>A</td><td colspan="2">2.5mM C13-glu 24h</td><td rowspan="2">20mM C13-glu 24h</td></tr>
<tr><td>B</td><td colspan="2">0.15mM C13-glu 24h</td></tr>
</table>

20250317coat

<table>
<tr><th></th><th>1</th><th>2</th><th>3</th></tr>
<tr><td>A</td><td colspan="2">2.5mM C13-glu 1h</td><td>2.5mM C12-glu 24h</td></tr>
<tr><td>B</td><td></td><td></td><td></td></tr>
</table>

2025年3月19日 星期三

T7-> T7
The plate which coated on 20250310 has much better cells
-> consider coating the plate over weekend with laminin

2025年3月20日 星期四

T7 -> BP T8

2025年3月22日 星期六

700uL BP T8 -> 800 uL BP T8

2025年3月24日 星期一

700uL BP T8 -> 800 uL BP T8
Thaw new iPSC into 1 6-well in StemFlex+ROCKI: hiPSC NIH wt-11 P16 Halo-Tag-TARDBP (N-term) 1/2 6-well 20240926 YST

2025年3月25日 星期二

change iPSC media into StemFlex only

2025年3月26日 星期三

- 700uL BP T8 -> 800 uL BP T8
- prepare media for GC-MS

Table4

	A	B	C	D	E	F	G	H	I
1			stock (mM)	targetconc. (mM)	dilution factor	final volume	added volume	media volume	
2	DMEM w/o glu	C12-glucose	1100	17.5	62.8571428571	2000	32	1968	
3			1100	0.15	7333.3333333333	7333	1	7332	
4		C13-glucose	100	17.5	5.7142857143	4743	830	3913	
5			100	0.15	666.6666666667				forgot how i prepared...
6	Neurobasal-A w/o glu	C12-glucose	1100	2.5	440	2640	6	2634	
7		C13-glucose	100	20	5	4000	800	3200	
8			100	2.5	40	6000	750 from 20mM	5250	
9			100	0.15	666.6666666667	4933	7	4926	

- split iPSC into 7 wells of 6-well

Table2

	A	B	C	D
1	conc. (K/mL)	survival	total (5mL)	K/well (into 7*6-well)
2	200	NA	1000	142.86

for keep

	1	2	3
A			
B			

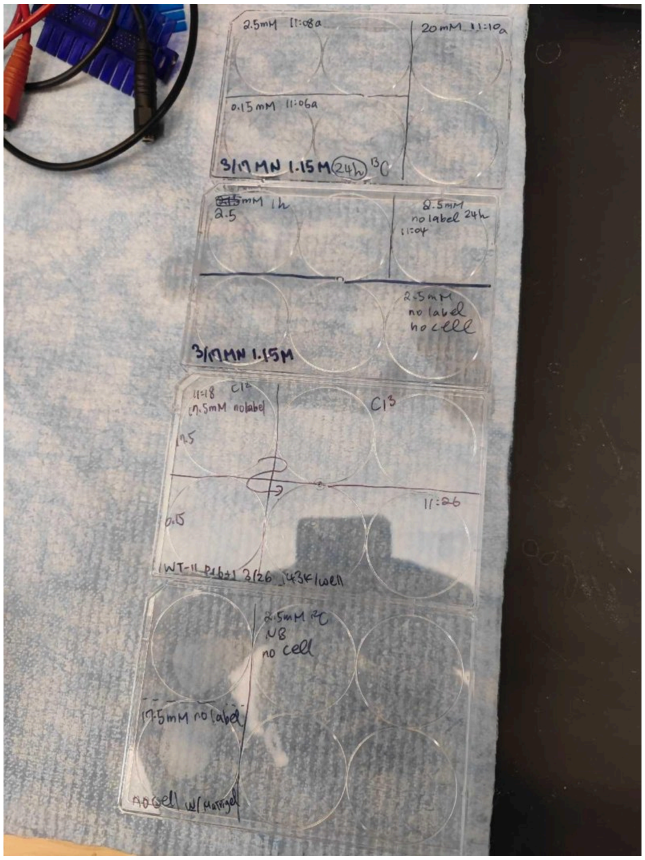
for GC

	1	2	3
A	2.5mM C12-glu in DMEM 24h	2.5mM C13-glu in DMEM 24h	
B	0.15mM C12-glu in DMEM 24h	0.15mM C13-glu in DMEM 24h	

2025年3月27日 星期四

treat cells (MN & iPSC 24h)

IMG_20250328_145840.jpg



2025年3月28日 星期五

GC extraction trial#2

Table1

	A	B	C	D	E	F	G	H	I
1		**tube name**	**cell/media type**	**glucose conc**	**label**	**treatment time (h)**	**replicate#**	**media sample**	**cell sample**
2	1	A	MN	0.15mM	C13	24	1	v	v
3	2	B	MN	0.15mM	C13	24	2	v	v
4	3	C	MN	2.5mM	C13	24	1	v	v
5	4	D	MN	2.5mM	C13	24	2	v	v
6	5	E	MN	20mM	C13	24	1	v	v
7	6	F	MN	20mM	C13	24	2	v	v
8	7	G	MN	2.5mM	C13	1	1	v	v
9	8	H	MN	2.5mM	C13	1	2	v	v
10	9	I	MN	2.5mM	C12	1	1	v	v
11	1	J	iPSC	17.5mM	C12	24	1	v	v
12	2	K	iPSC	17.5mM	C13	24	1	v	v
13	3	L	iPSC	17.5mM	C13	24	2	v	v
14	4	M	iPSC	0.15mM	C12	24	1	v	
15	5	N	iPSC	0.15mM	C13	24	1	v	
16	6	O	iPSC	0.15mM	C13	24	2	v	
17	coating only controls	coat1	Neurobasal-A w/ PLO+laminin (old)	2.5mM	C12	24	1	v	v
18		coat2	Neurobasal-A w/ PLO+laminin(new)	2.5mM	C12	24	1	v	v
19		coat3	DMEM+ROCKI w/ matrigel	17.5mM	C12	24	1	v	v
20	media controls	M1	DMEM	0.15mM	C12			v	
21		M2	DMEM	17.5mM	C12			v	
22		M3	DMEM	0.15mM	C13			v	
23		M4	DMEM	17.5mM	C13			v	
24		M5	Neurobasal	0mM				v	
25		M6	Neurobasal	2.5mM	C13			v	
26		M7	DMEM	0mM				v	

2025年4月11日 星期五

Analysis

- need to know what the m/z is for each metabolite (sometime they have smaller fragments)
- choose the time when it comes out and try to find a consistent peak
- If we don't know where it is, we need to run a sample with the internal standard only

Result:

1. machine seemed stuck, so the metabolite coming out later didn't have good signal (like histidine)

- medium samples:
    - DMEM has no alanine and proline
    - succinate and malate has weird peak
    - fumerate has a lot of background
    - glutamine may degrade into glutamate spontamenously
        - 0.15%/day at 4C
        - ref: https://www.sciencedirect.com/science/article/pii/026156149190037D#:~:text=The%20degradation%20rate%20of%20L%2Dglutamine%20in%20the%20intravenous%20solutions,no%20associated%20formation%20of%20glutamate.
    - There is always a histidine with C12??
    - tyrosine didn't run through in internal standard
- iPSC:
    - pyruvate, lactate, alanine have similar pool: similar to cancer cell
        - 0.15mM glucose might survive around 1-2 h
        - search how much glucose iPSCs can tolerate
    - succinate: 20% M2 in 24h (have similar pool as pyruvate, but run slower)
    - fumarate: have weird M3 and M2 signal despite there are many M4 signal (maybe background)
        - iPSCs might run a reverse cycle
    - serine should be good
    - a-KG may not a stable substrate: still could not be detected
    - Malate: still has M3 signal -> might because of the reverse cycle from oxaloacetate. Media has much exorted malate (interesting)
    - proline: from a-KG
    - glutamate: looks okay

Table5

	A	B	C	D
1				
2	pyruvate			
3	lactate			
4	alanine			
5	glycine			
6	valine			
7	leucine			
8	isoleucine			
9	succinate			
10				
11				
12				

- coating sample:
    - DMEM+matrigel & neurobasal-A+coating (new): somehow has a big malate peak --> is the malate exist in coating reagent
    - can run coating reagent only
- MN:
    - 1h has label!
    - 24h pattern is pretty close to the data we had last time

- to do with Pacold lab: one full 6-well plate of iPSCs